

Chapter 10

Chirality

Learning Objectives

- Isomerism: structural isomer, stereoisomer
 - Enantiomer and diastereomer
- Chirality
- Chiral center
- Absolute configuration
- Meso compound

Part 1

Classification of Isomerism

STEREOCHEMISTRY

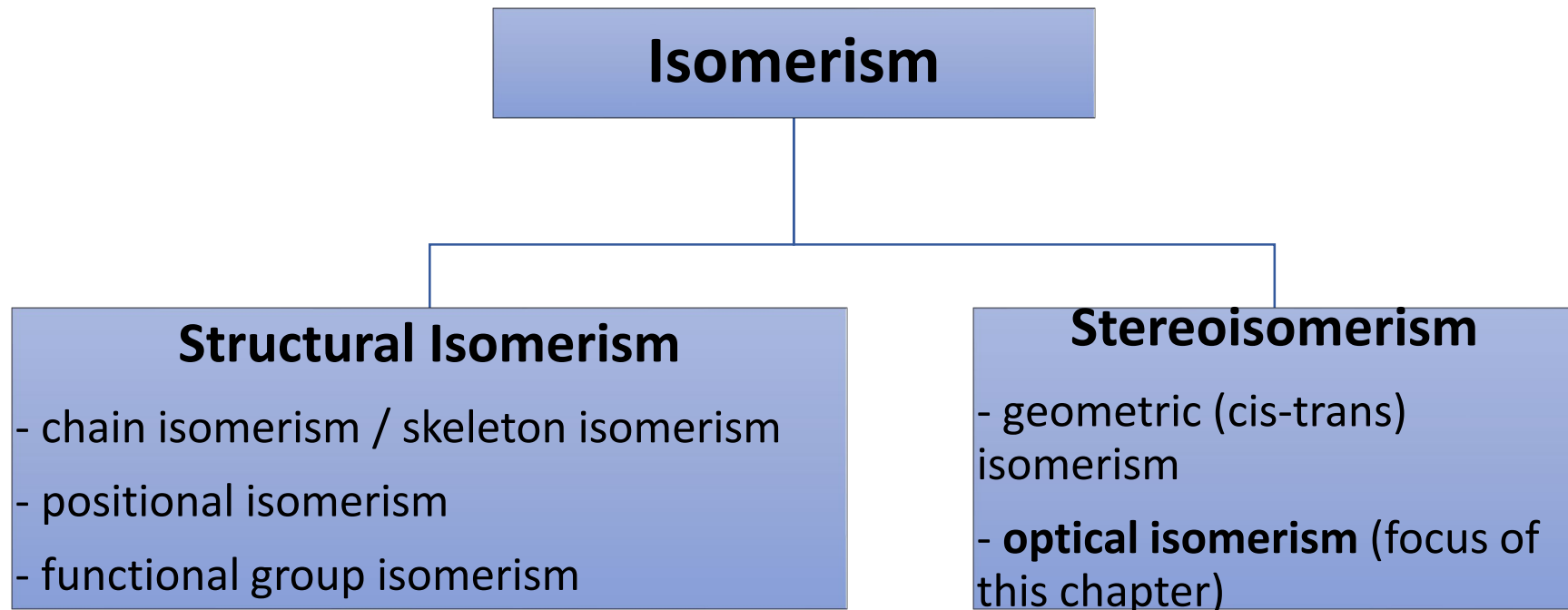
THE ARRANGEMENT OF ATOMS IN SPACE

Stereochemistry is the study of the structure of the molecules in their three-dimensional geometry.

Molecules that have the same connectivity but different arrangement of atoms in space are called stereoisomers.

Classification of Isomerism

- **Isomerism:** Occurs when two or more compounds have the same molecular formula but different arrangement of the atoms in the molecules.



Structural Isomerism vs Stereoisomerism

Structural Isomerism

- structural isomers have the same **molecular** formula but different **structural** formula
- arises from the different ways in which atoms can join together to form molecules
- atoms are **linked in different sequences** and in different ways

Stereoisomerism

- stereoisomers have same **molecular** and **structural** formula, but *different arrangement of their atoms in space*
- arises from the *different ways* in which various groups in the molecule can *arrange in space*
- atoms are linked in the same sequence and in the same way but *differ in their arrangement in space*

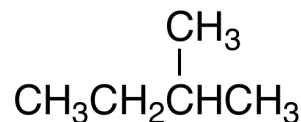
Structural Isomerism

Chain isomerism – different arrangement of carbon atoms in a chain

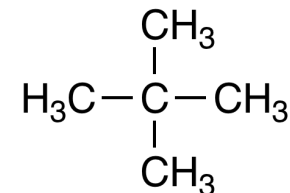
eg. isomers of pentane (molecular formula C_nH_{2n+2})



pentane
(*n*-pentane)



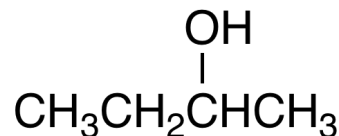
2-methylbutane
(*iso*-pentane)



2,2-dimethylpropane
(*tert*-pentane)

Positional isomerism – different positions of functional group in the carbon chain

eg. 1-butanol and 2-butanol



Nomenclature of
alcohol/ether is not
required for this course.

Functional group isomerism – different functional groups

eg. 1-butanol and ethoxyethane (both $C_4H_{10}O$)



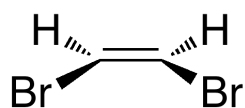
alcohol



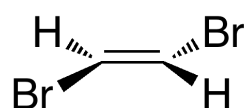
ether

Stereoisomerism

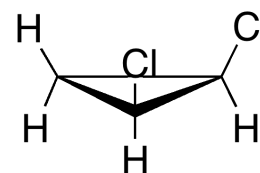
Geometric isomerism – arises when rotation about a bond is restricted



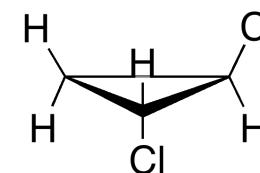
cis-isomer



trans-isomer



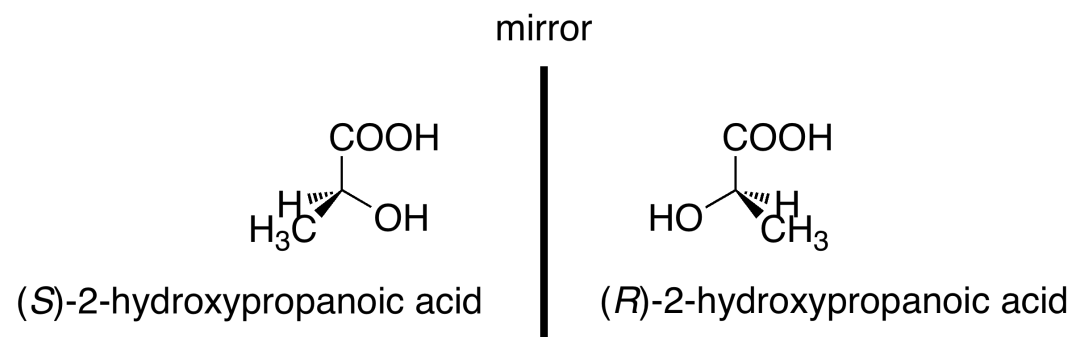
cis-isomer



trans-isomer

Optical Isomerism (Enantiomerism)

- Arises from lack of symmetry of the molecule
- Molecule is able to rotate the planar-polarized light → optical isomerism



Part 2

Chirality

Handedness

- Is this a left hand or a right hand?



- Is there anything wrong of this ancient Egyptian painting?

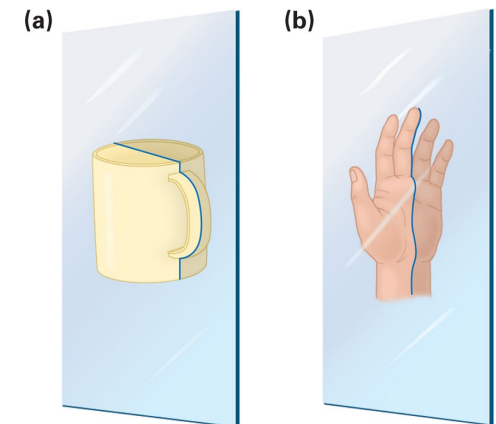
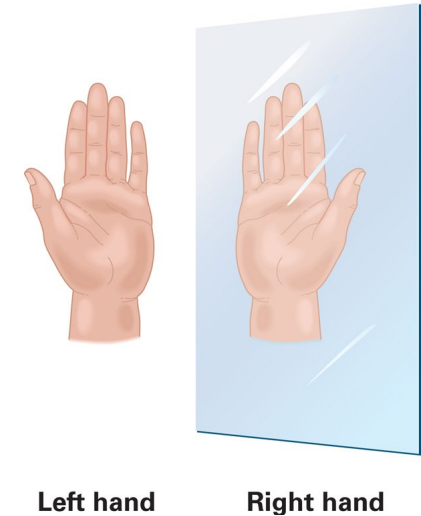


- Our two hands – are they the same or different? Think about gloves.

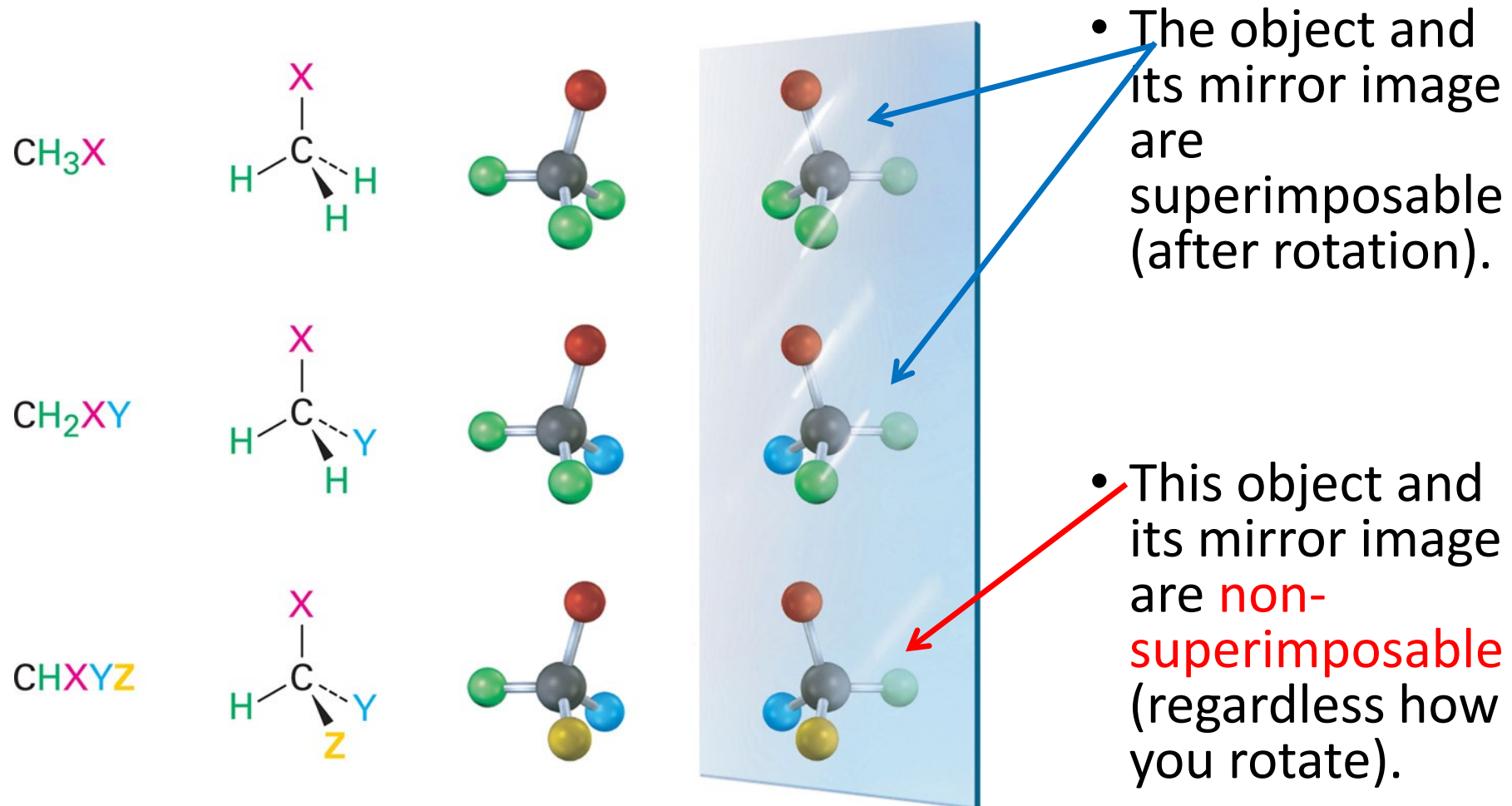


Our hands are chiral

- Two hands are non-superimposable mirror images.
- Each hand does not have a plane of symmetry.
- The term “chiral” or “chirality” is derived from the Greek word $\chi\epsilon\iota\rho$ (*kheir*), “hand”.
- A chiral object is **non-superimposable** with its mirror image.

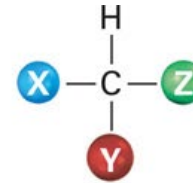
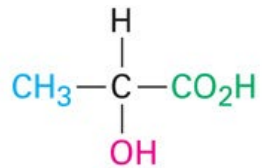


Molecules can be chiral too

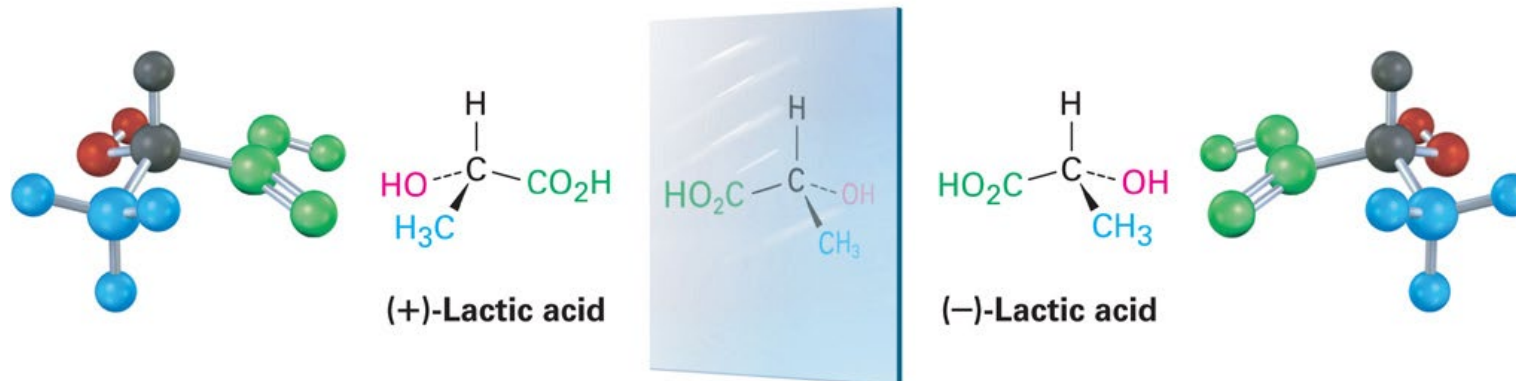


Enantiomers

- Enantiomers: a pair of **mirror images** that are **non-superimposable**.

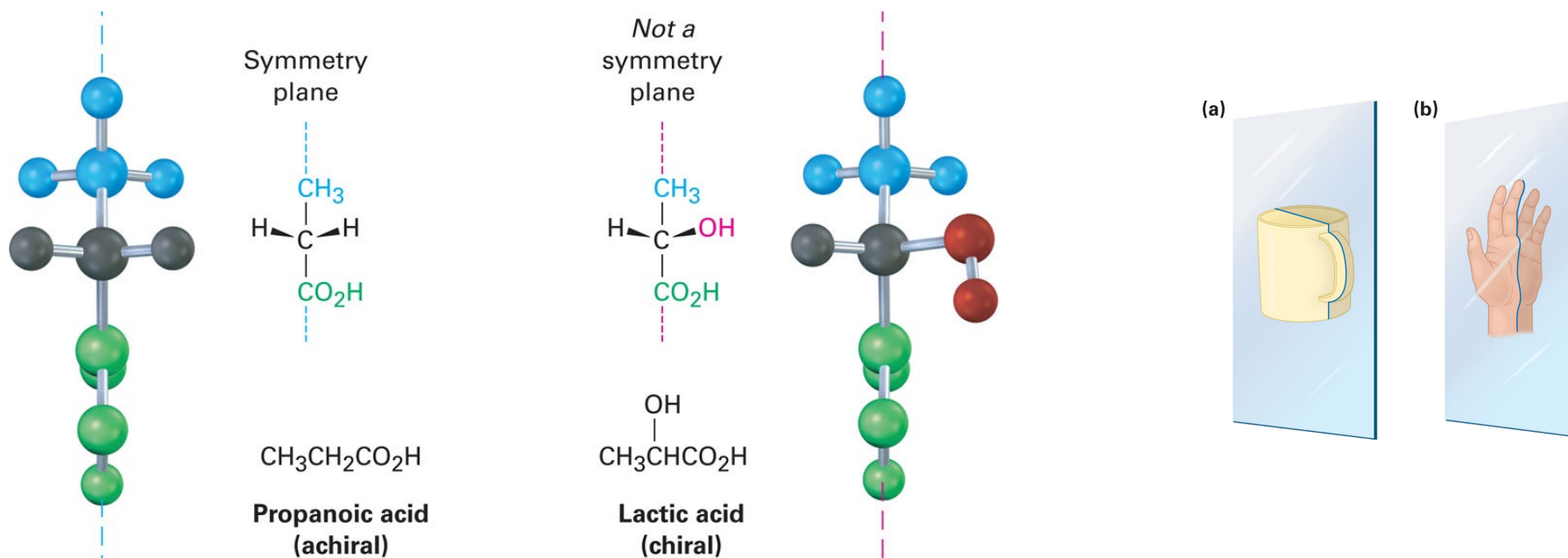


Lactic acid: a molecule of the general formula CHXYZ



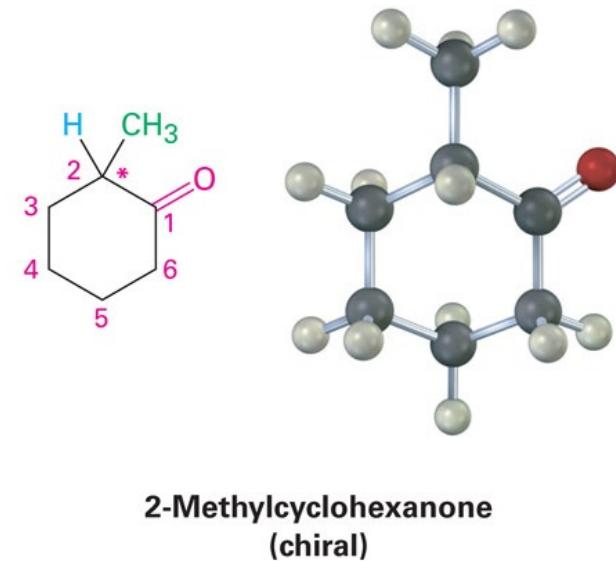
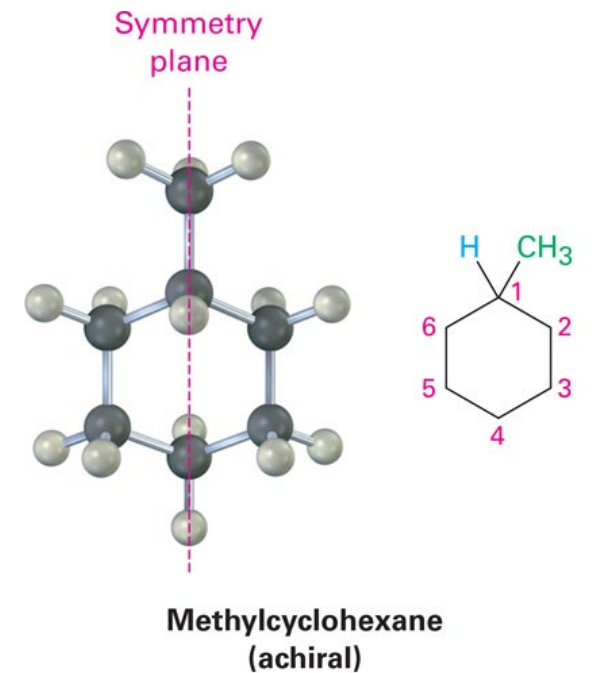
Plane of Symmetry

- Chiral molecules/objects have no plane of symmetry.
- Achiral molecules/objects have plane of symmetry.

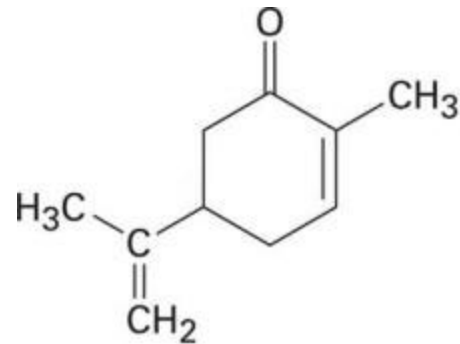


Chiral center

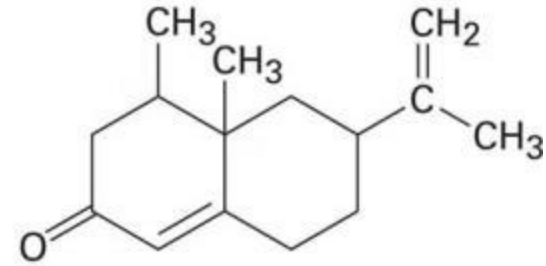
- Chiral center is the most common cause of chirality, but it is not the only cause.
- A chiral center (also called stereocenter, asymmetric center or stereogenic center) is a **tetrahedral** atom bonded to four **different** groups.
- The chiral center can be various elements, but most common is an sp^3 carbon. [sp^2 -C and sp -C are never chiral centers.]
- Chiral centers are usually marked with an asterisk, *



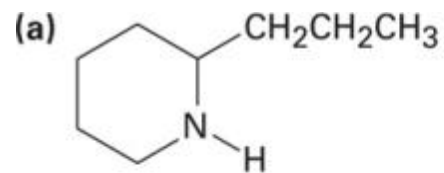
Identify Chiral Centers



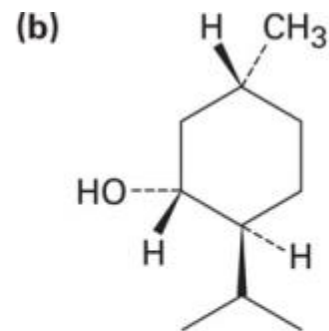
Carvone (spearmint oil)



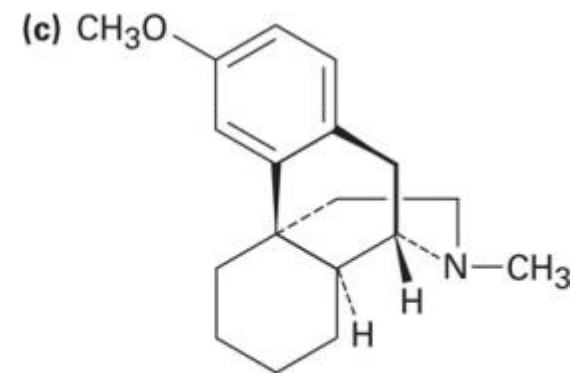
Nootkatone (grapefruit oil)



**Coniine
(poison hemlock)**



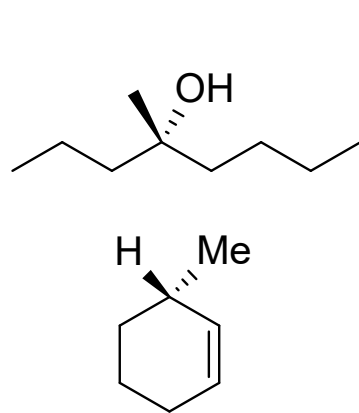
**Menthol
(flavoring agent)**



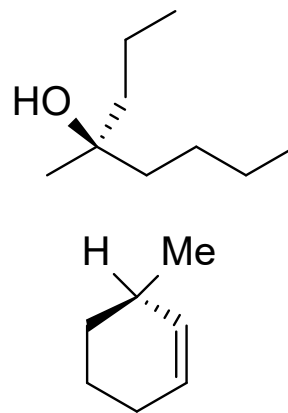
**Dextromethorphan
(cough suppressant)**

How to draw chiral molecules

- Solid line: the bond is on the (paper) plane
- Bold/wedged line: the bond is out of the (paper) plane and pointing towards the observer
- Dashed/dotted line: the bond is out of the (paper) plane and pointing away from the observer
- By convention, the ring or longest carbon chain is on the solid lines



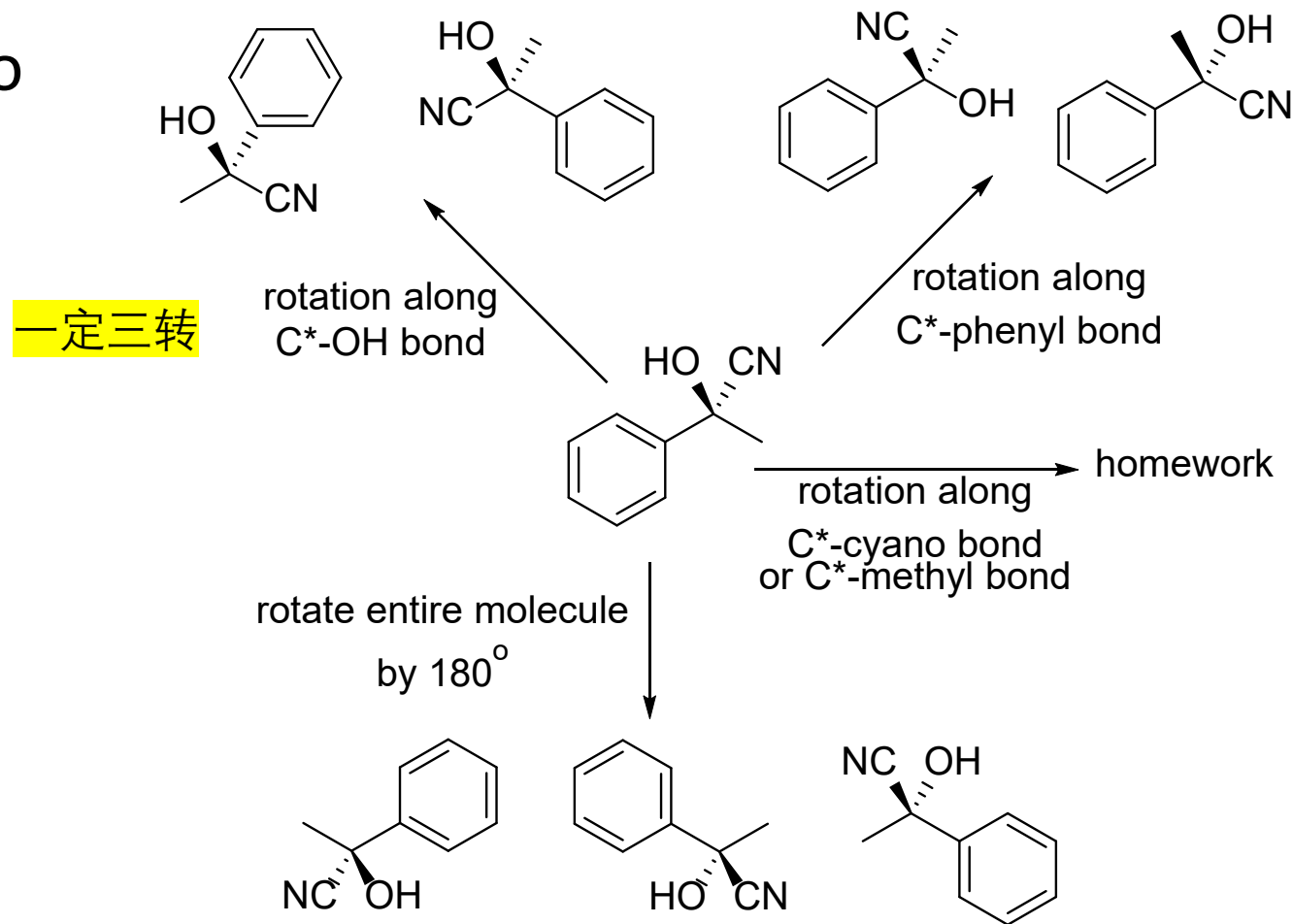
proper drawing
following convention



nothing wrong,
but unconventional

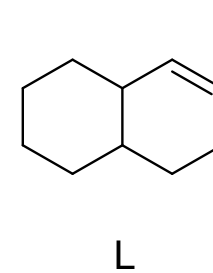
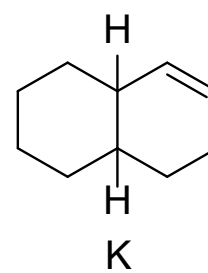
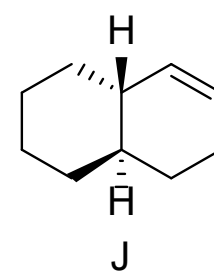
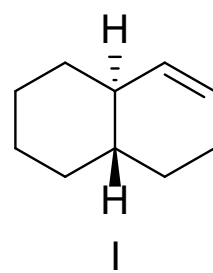
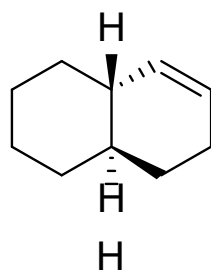
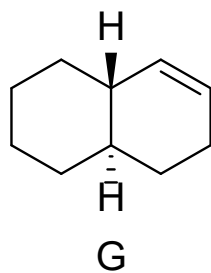
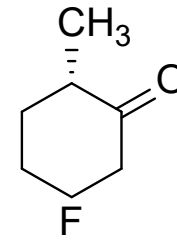
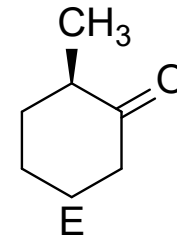
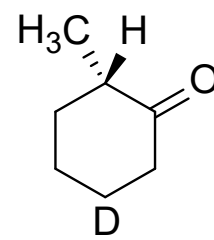
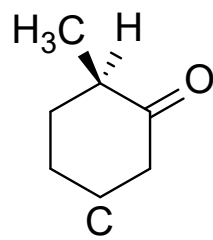
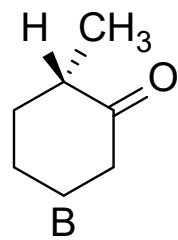
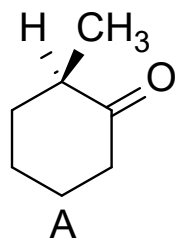
Multiple drawings of the same enantiomer

- Multiple ways to draw the same enantiomer:



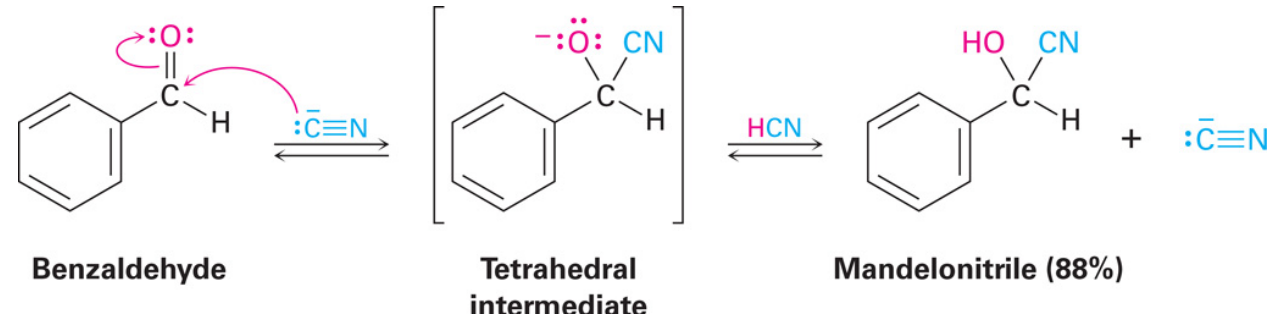
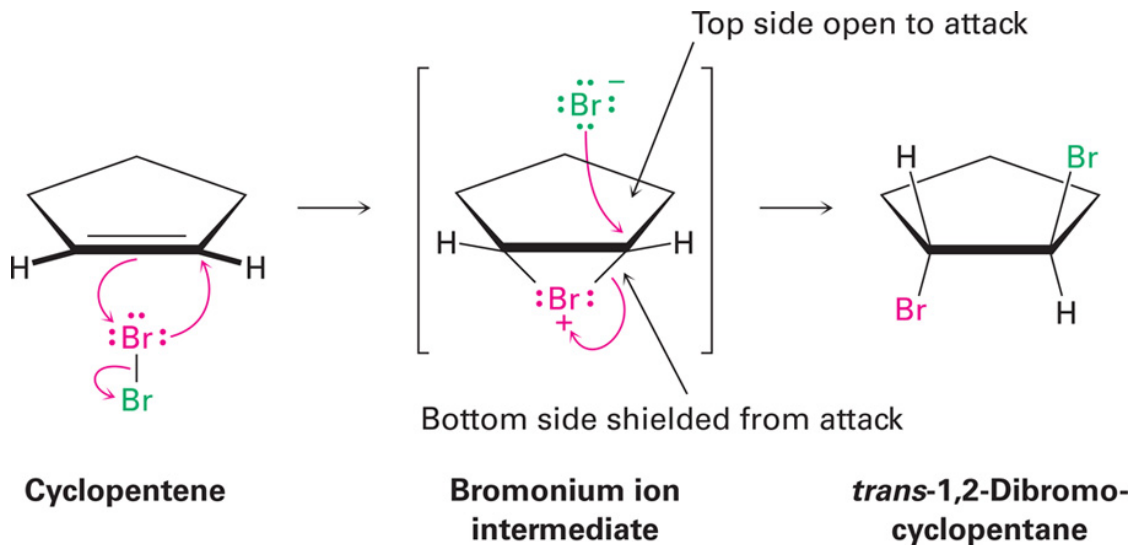
Exercise (1)

- Which drawings are identical?



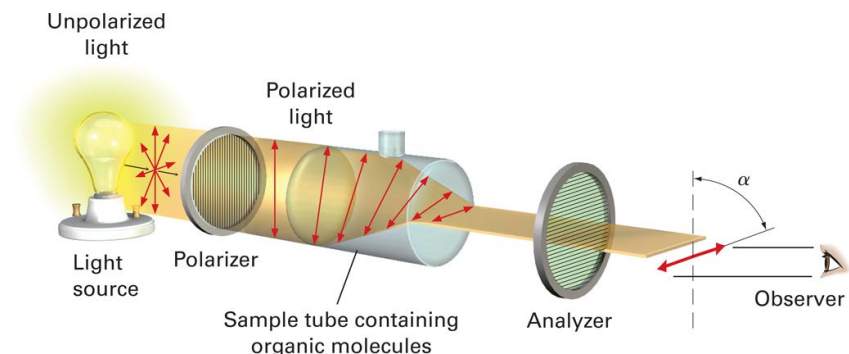
Racemic mixture

- Racemic mixture (or racemate) is a 50:50 mixture of two enantiomers
- When there is no chiral reactant/reagent/catalyst/solvent, the two enantiomers have equal probability to form → racemic mixture is formed
- The symbol is ($\pm$)



Optical Activity

- A pair of enantiomers have identical physical properties (e.g. mp, bp and solubility) and chemical properties, except:
 - Enantiomers will rotate plane-polarized light to opposite directions
 - One enantiomer will rotate the light to the left (counterclockwise) → levorotatory or (-)-enantiomer
 - The opposite enantiomer will rotate the light to the right (clockwise) → dextrorotatory or (+)-enantiomer
 - The magnitude of rotation is the same, just opposite direction
- A single enantiomer (+ or -) is optically active; racemic mixture ($\pm$) is optically inactive.



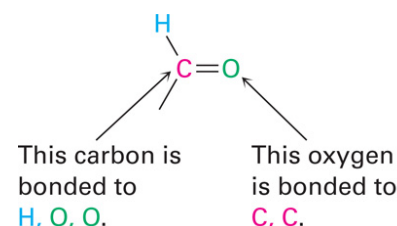
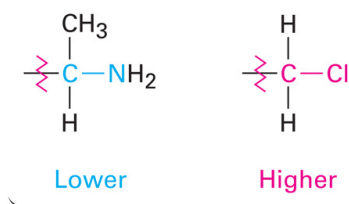
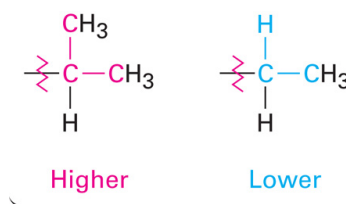
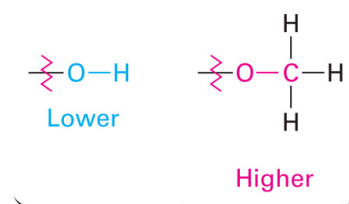
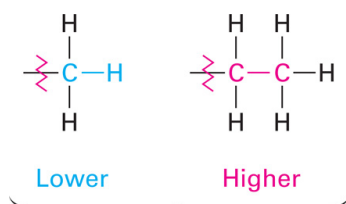
Absolute configuration

- Step 0: Identify the chiral center and find out the four different groups attached to the chiral center.
- Step 1: Rank the four groups by Cahn-Ingold-Prelog (CIP) rule: the group with highest priority is ranked 1, lowest priority is ranked 4
- Step 2: Orient the molecule such that the lowest-priority group (4) is **pointing away** from the observer
- Step 3: Count the direction of priority group $1 \rightarrow 2 \rightarrow 3$
 - If $1 \rightarrow 2 \rightarrow 3$ gives a clockwise direct $\rightarrow$ R configuration
 - If $1 \rightarrow 2 \rightarrow 3$ gives a counterclockwise direct $\rightarrow$ S configuration

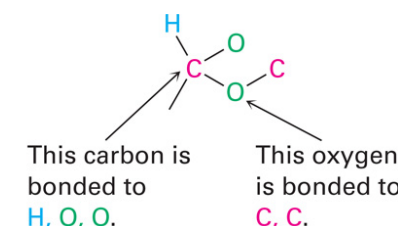
Recap of the CIP rule

Atomic number 35 17 16 15 8 7 6 (2) (1)
 Higher ranking Br > Cl > S > P > O > N > C > ²H > ¹H Lower ranking

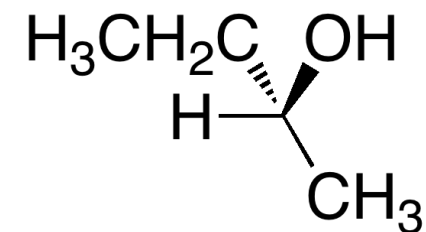
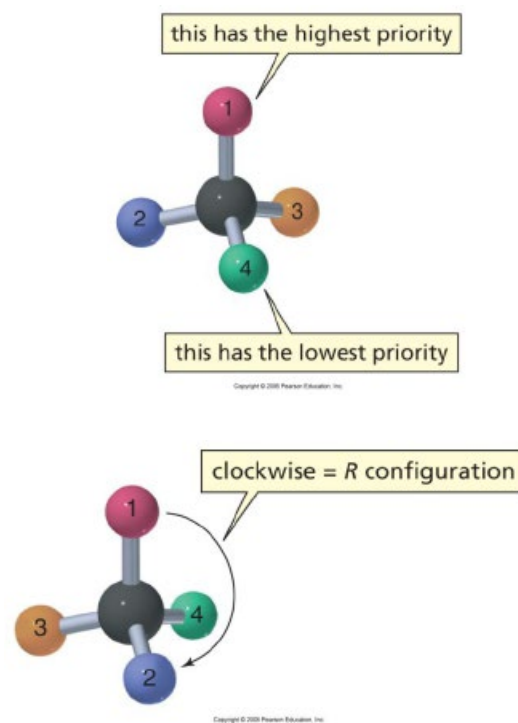
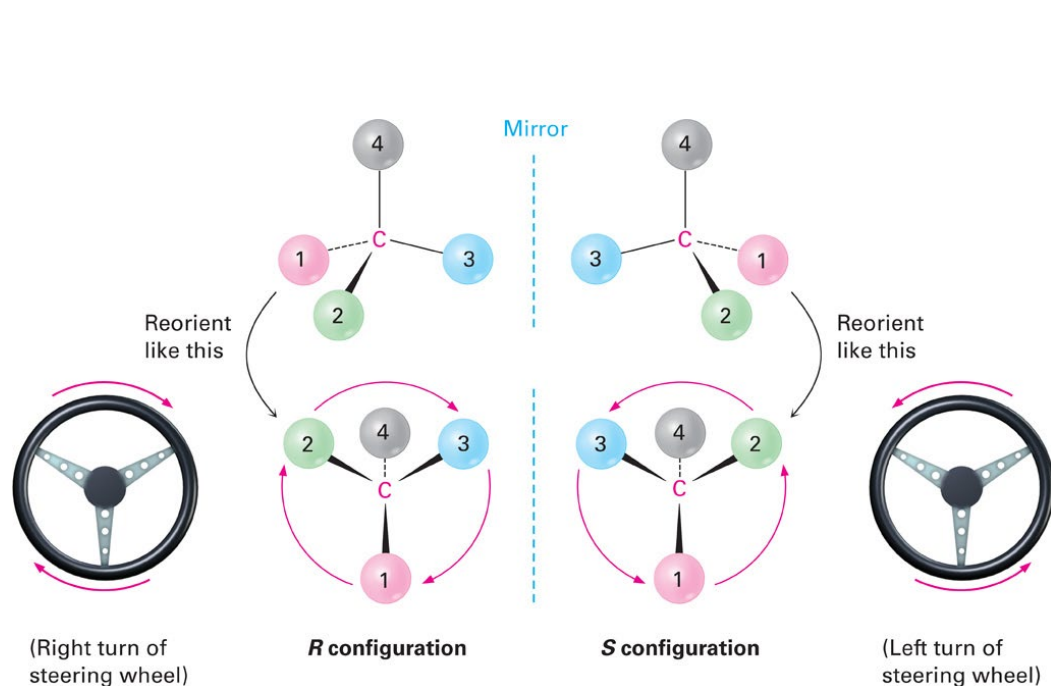
1. Compare the directly attached atom by atomic number → higher atomic number (higher atomic mass for isotopes) has higher priority
2. If the directly attached atom (level I) is identical → compare the atoms further attached (level II); If level II are still identical → compare level III, and so on
3. Multiple-bonded atom is considered as single-bonded atom multiple times



is equivalent to



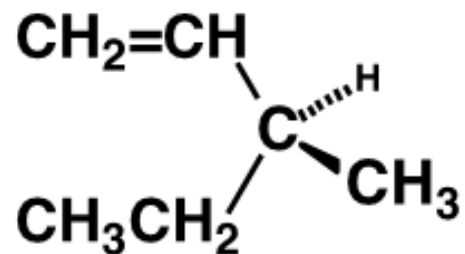
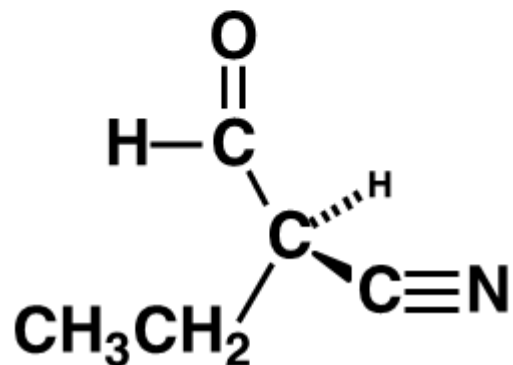
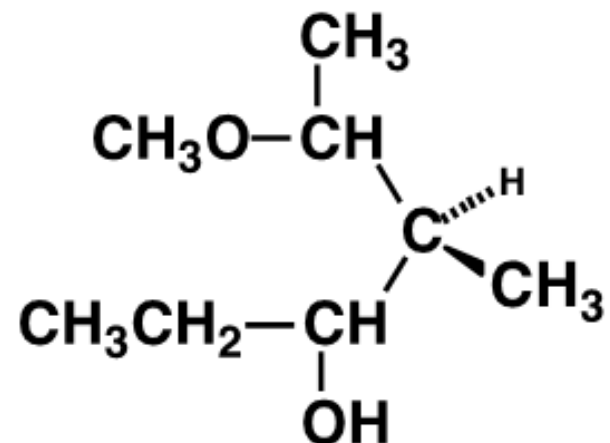
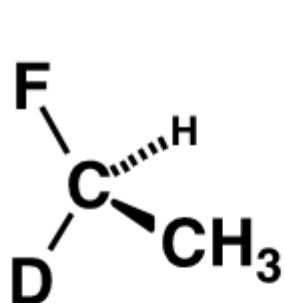
Assignment of Absolute Configuration



Absolute configuration (R/S) is unrelated to optical activity (+/-).

Exercise

- Assign absolute configuration for each chiral center.

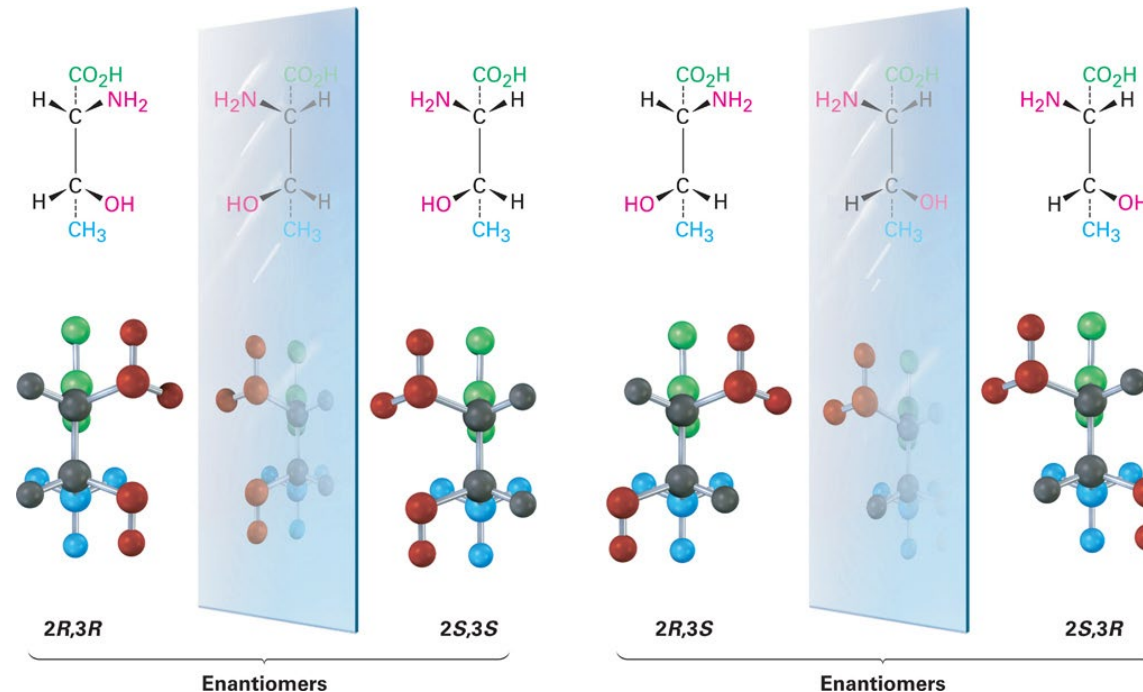


Exercise

- Draw (S)-3-chloropent-1-ene.

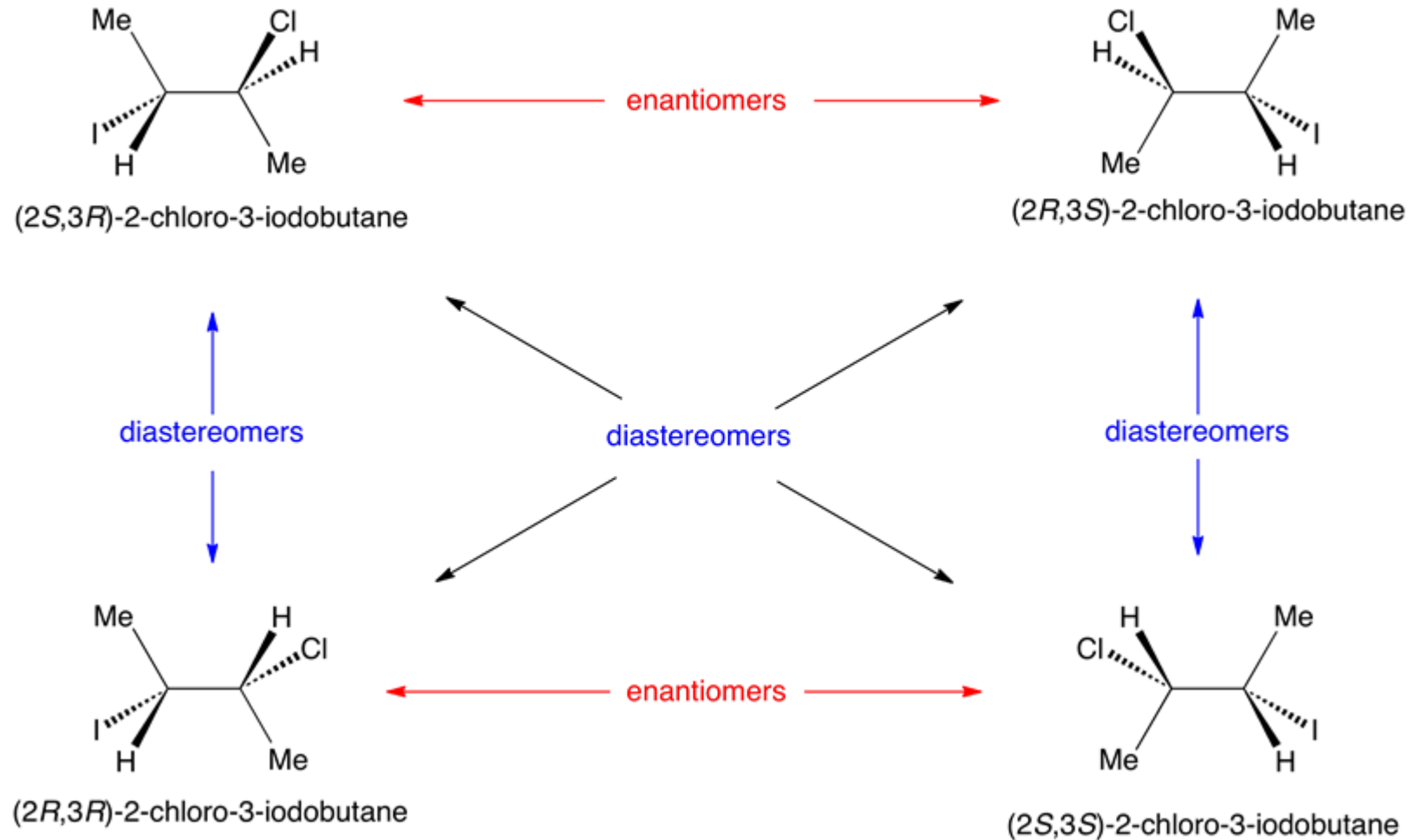
Diastereomers

- For a molecule with n chiral centers, the **maximum possible** number of stereoisomers is 2^n (each chiral center gives R & S)
- Two pairs of enantiomers: 2R,3R & 2S,3S; 2R,3S & 2S,3R
- Diastereomers: non-superimposable & non-mirror-images (e.g. 2R,3R vs 2R,3S)



Example of Diastereomer/Enantiomer

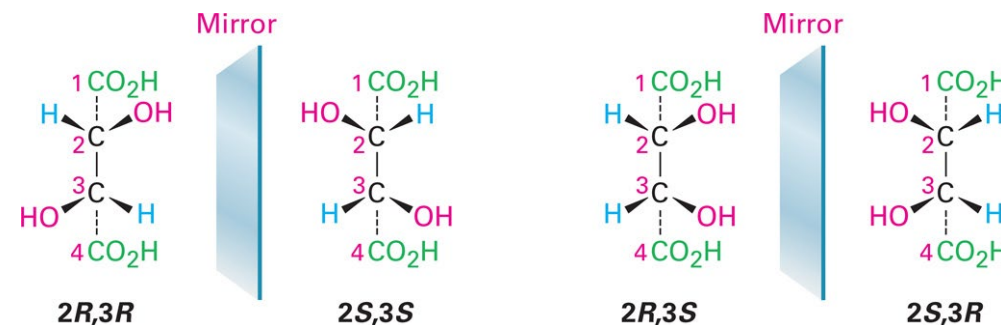
2-chloro-3-iodobutane



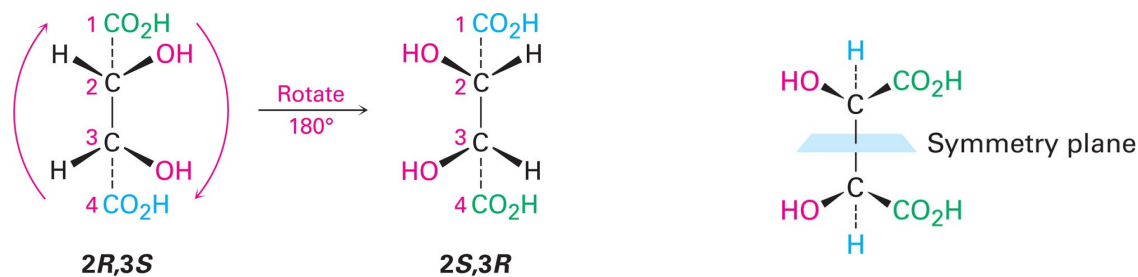
- Enantiomers have **identical** physical and chemical properties, except the different direction to rotate planar polarized light.
- Diastereomers (may) have **different** physical and chemical properties.

Meso Compound

- Are there $2^2=4$ stereoisomers of tartaric acid?

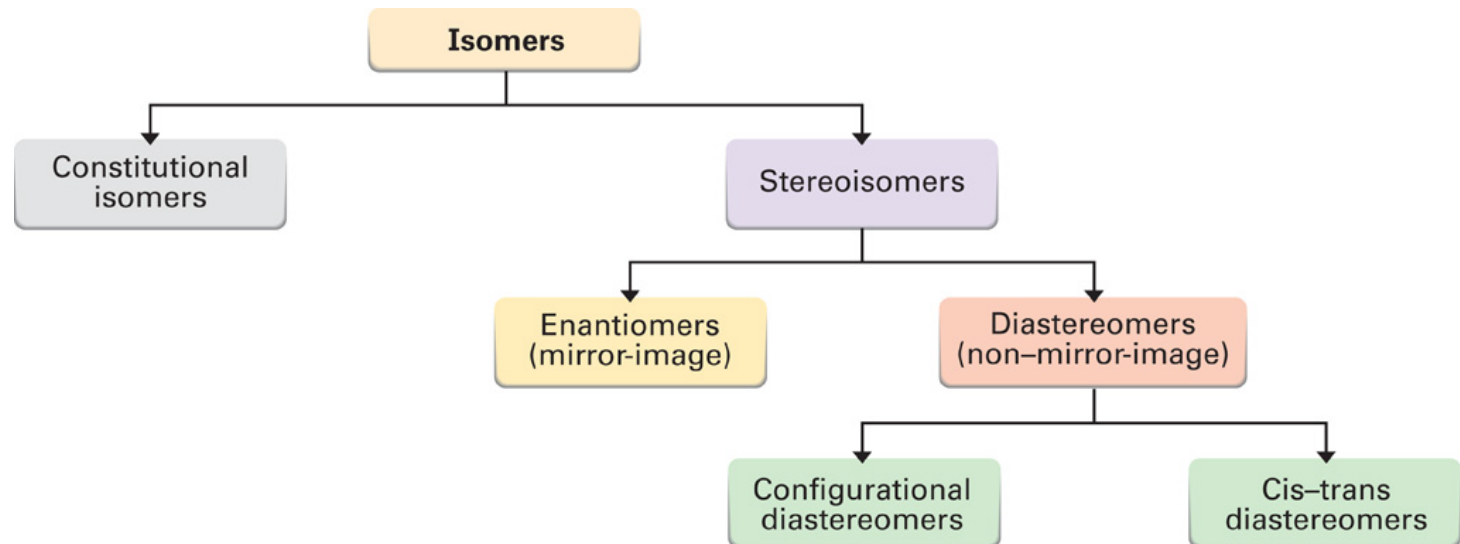


- No! 2R,3S and 2S,3R are actually identical $\rightarrow$ meso compound
- Meso has an internal plane of symmetry. It is an “achiral” molecule but there are chiral centers within the molecule.
- Meso compound has different physical and chemical properties.

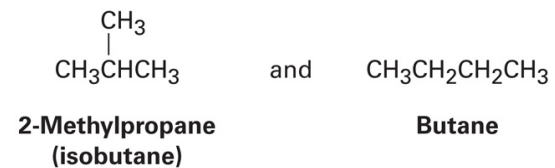


Stereoisomer	Melting point (°C)	$[\alpha]_D$	Density (g/cm ³)	Solubility at 20 °C (g/100 mL H ₂ O)
(+)	168–170	+12	1.7598	139.0
(-)	168–170	-12	1.7598	139.0
Meso	146–148	0	1.6660	125.0

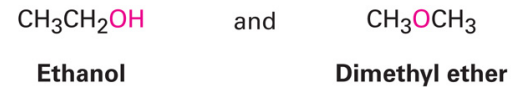
Isomerism revisited



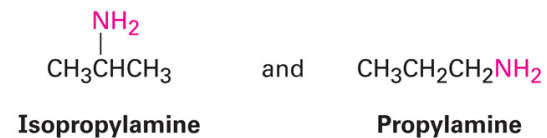
Different carbon
skeletons
 C_4H_{10}



Different functional
groups
 C_2H_6O

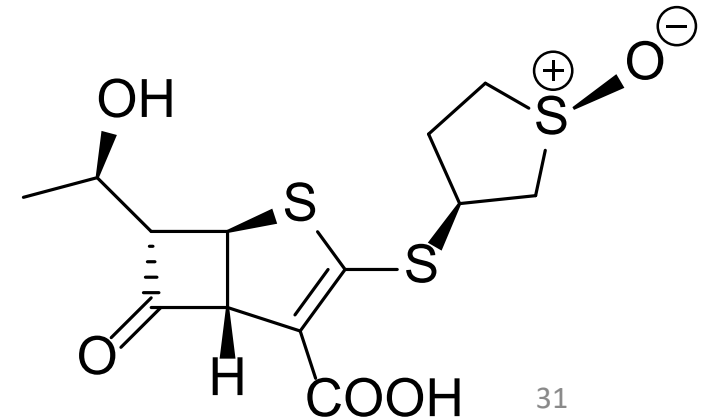
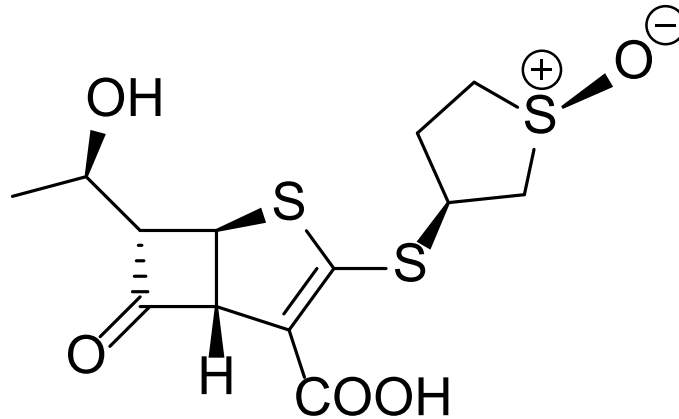
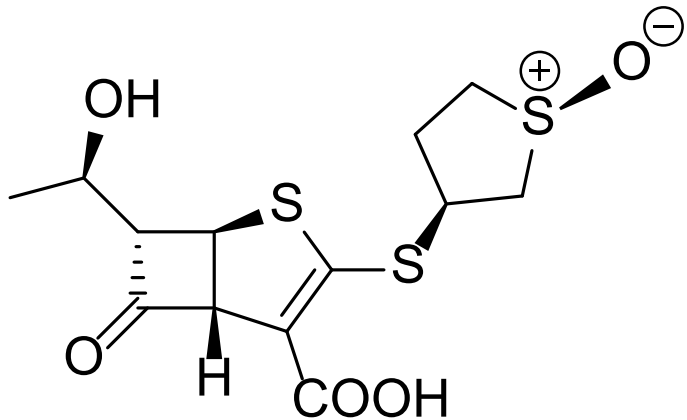


Different position of
functional groups
 C_3H_9N



Exercise

- Label each chiral center
- Indicate the priority of the four groups attached to this chiral center
- Determine the absolute configuration of this chiral center

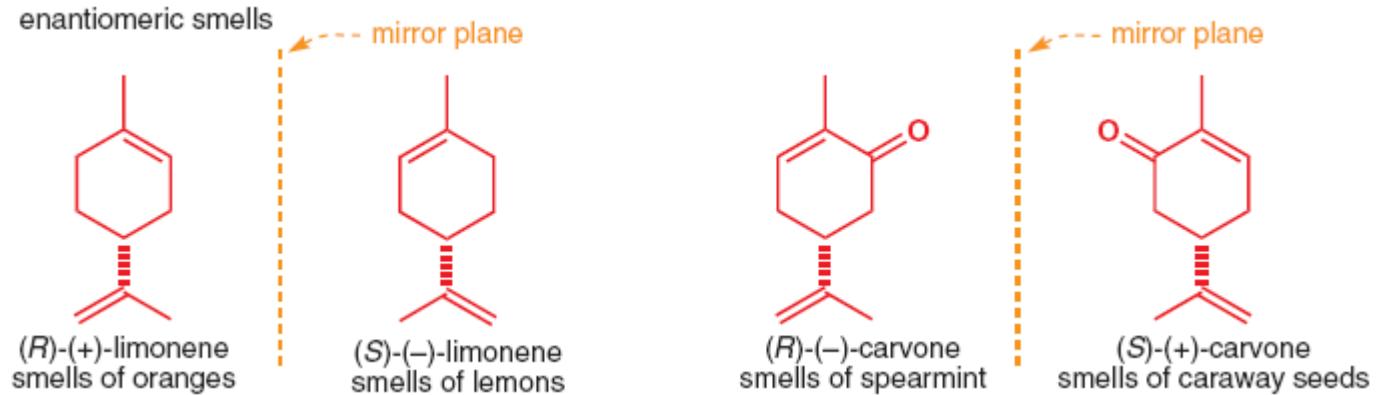


Part 3

Chirality in Nature and Science

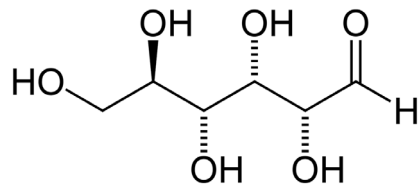
Why is Chirality important?

- Chirality in nature

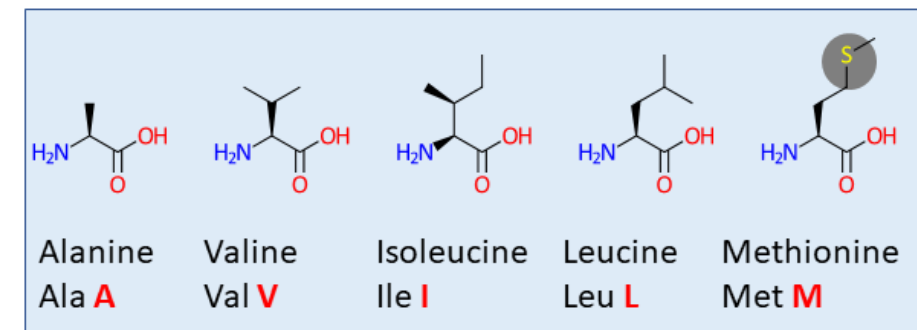


- All living systems are chiral

- Amino acids
- Sugars
- DNA & RNA

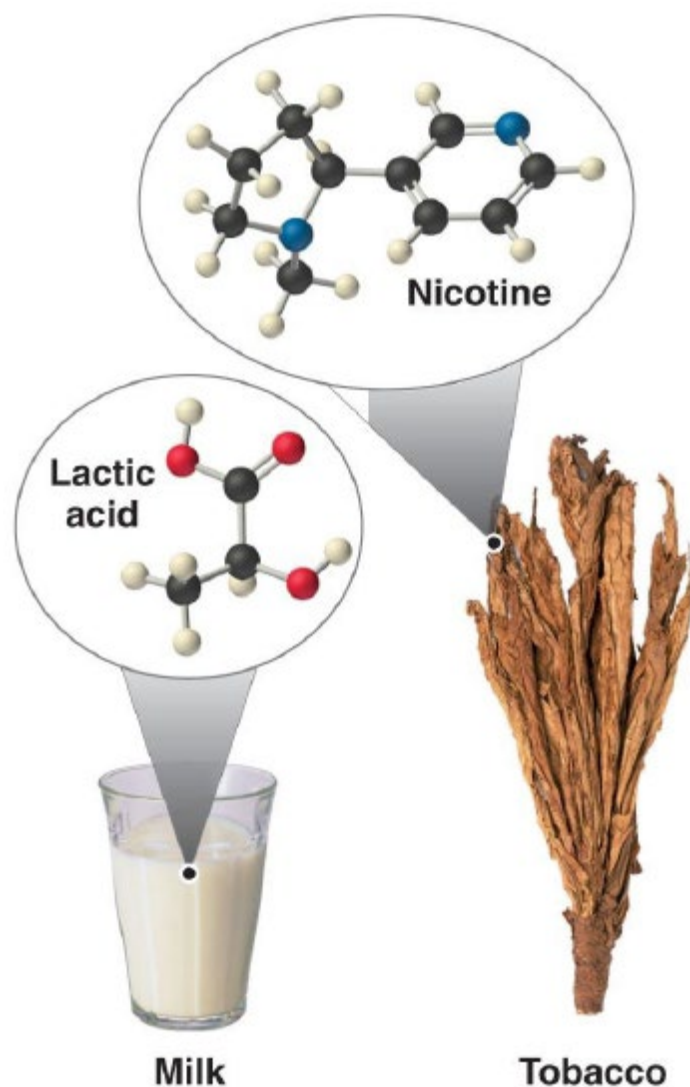


Each sugar (carbohydrate) is
a single enantiomer

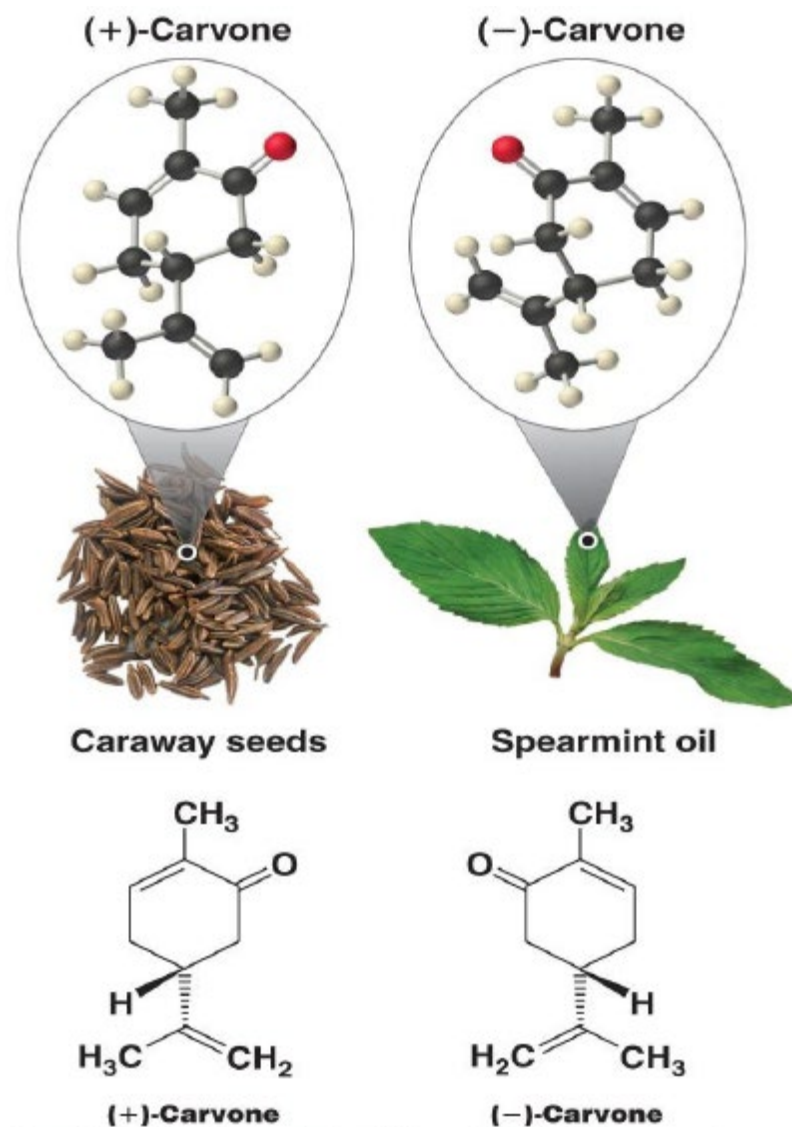


All natural amino acids are (R)-isomer, but not (S)

Some examples of molecules containing a chiral center or asymmetric carbon atom



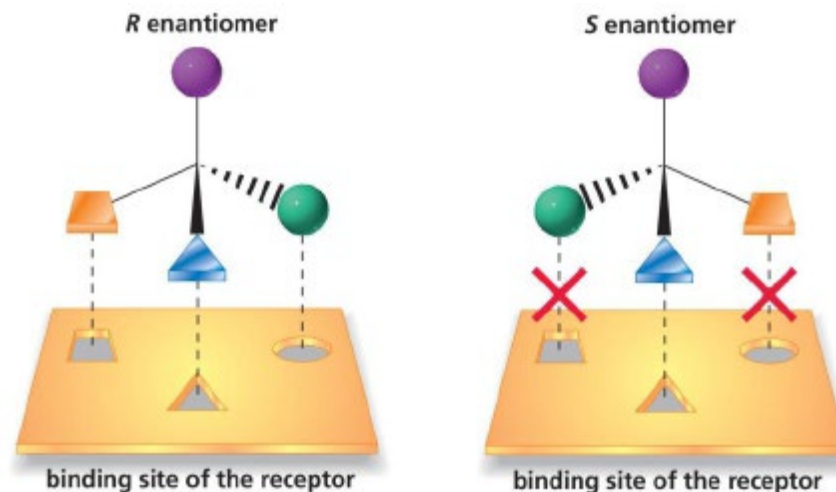
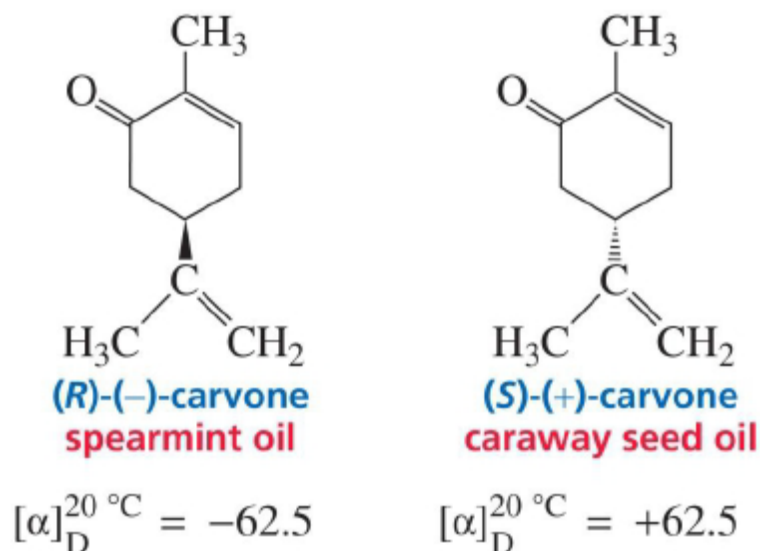
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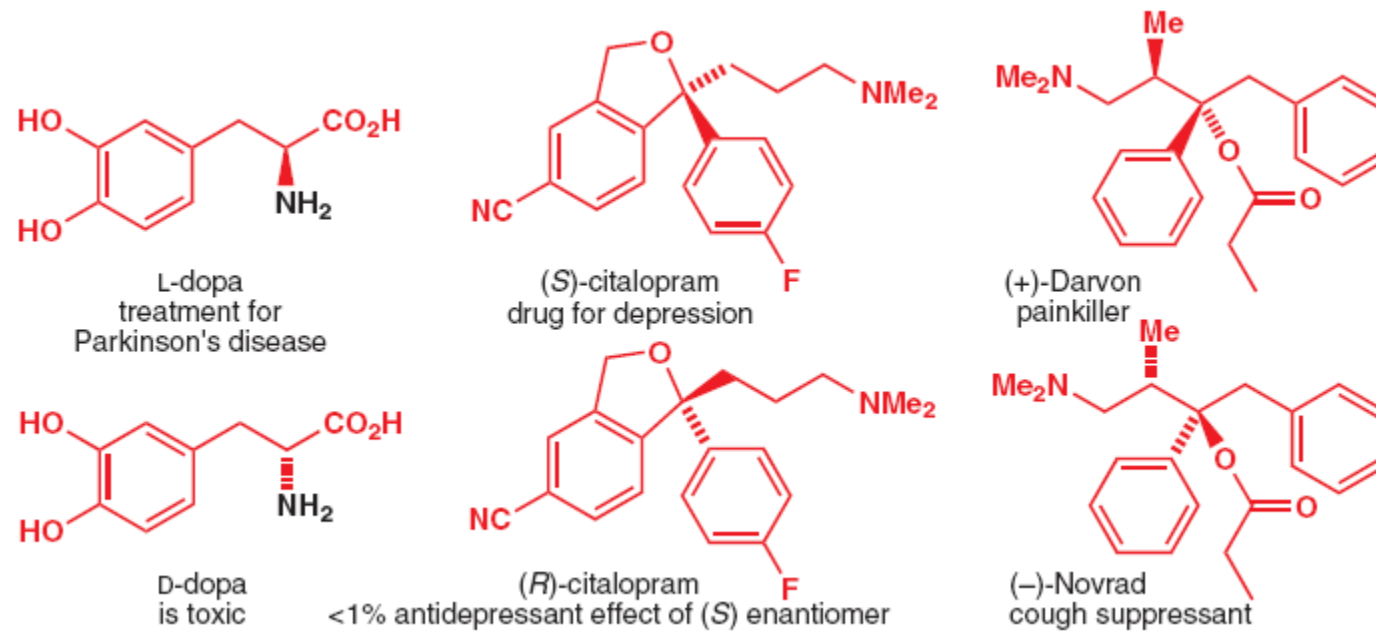
Why opposite enantiomers smell different?

- A receptor is a protein that binds a particular molecule.
- A receptor, being chiral, will bind one enantiomer better than the other.
- Hence two enantiomers have different physiological properties.



Our body is a chiral environment

- Opposite enantiomer may interact with our body in different ways.
- 汝之蜜糖 彼之砒霜



Thalidomide Tragedy

- Developed by the German pharmaceutical company Chemie Grünenthal in the 1950s. Marketed to pregnant women as a remedy for morning sickness and nausea.
- Thalidomide caused severe birth defects, particularly limb malformations (commonly referred to as "phocomelia"), where babies were born with shortened or absent limbs. It was later found that the (S)-enantiomer has anti-nausea effect, but (R)-enantiomer is teratogenic. 致畸 止吐
- US FDA has since required single enantiomer, but not racemic mixture, to be tested separately and marketed as medicine.
- Chirality is crucial for pharmaceutical industry.